

The Arithmetic of Elliptic Curves

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Preface

Chapter 1

Lecture Notes

1.1 26/06/15 - Introduction I: Diophantine Equations, Plane Curves, and Bachet's Example

One of the oldest and most natural questions in mathematics goes back at least to the 3rd century CE, and probably much more. This question is the study of solutions to *Diophantine equations*, named after Diophantus of Alexandria, who “formulated and solved many such problems” ([1, p. xv]).

A *Diophantine equation* is an equation of the form

$$f(x_1, \dots, x_n) = 0$$

where $n \in \mathbb{Z}_{\geq 1}$, and $f \in \mathbb{Q}[x_1, \dots, x_n]$ is a polynomial in n variables with rational coefficients. To *solve it* means to find all rational solutions $(x_1, \dots, x_n)$ to the equation, although what it means to “find” all solutions can indeed be quite subjective. What is usually meant and hoped for is some sort of complete description or parametrization of such solutions. Variations of this problem ask for solutions over a ring R to similar equations defined over R instead of $\mathbb{Q}$ for various rings R such as $\mathbb{Z}$, a number field K , or its ring of integers $\mathcal{O}_K$. Of course, there is a cool interaction between solutions over $\mathbb{Z}$ and $\mathbb{Q}$; see 1.1.3 for one example.

There are (at least) two measures of the complexity of a given Diophantine equation: the number n of variables, and the degree $d \in \mathbb{Z}_{\geq 1}$ of the polynomial f . Let's study some examples for small n and d .

Example 1.1.1 (Single-Variable Equations). The case of $n = 1$ corresponds to equations of a single variable $x = x_1$.

- (a) When $d = 1$, we are looking at linear equations in one variable, which are of the form $ax + b = 0$ for some $a, b \in \mathbb{Q}$ with $a \neq 0$. I trust the reader knows how to solve such equations.
- (b) When $d = 2$, we are looking at solving quadratic equations over $\mathbb{Q}$, which look like $ax^2 + bx + c = 0$ for $a, b, c \in \mathbb{Q}$ with $a \neq 0$. Our insistence on asking for rational solutions means that there are essentially four things that can happen, depending on the discriminant $\Delta = b^2 - 4ac$; these are:
 - (i) There is a unique root of “multiplicity two”; this happens iff $\Delta = 0$ (e.g., $x^2 - 4x + 4 = 0$).
 - (ii) The two distinct roots are non-real complex conjugates; this happens iff $\Delta < 0$ (e.g., $x^2 - 5x + 5 = 0$).
 - (iii) The two distinct roots are irrational but real Galois conjugates; this happens iff $\Delta > 0$ but Δ is not a perfect square (e.g., $x^2 - 5x + 7 = 0$).
 - (iv) The two distinct roots are both rational; this happens iff Δ is a perfect square (e.g., $x^2 - 5x + 6 = 0$).
- (c) When $d = 3, 4$, more complicated formulae exist for the solutions, but these are not always helpful. Sometimes we can use elementary calculus to estimate the number of real roots (see, e.g., 2.1.1). When $d \geq 5$, a famous theorem says that no such formula in terms of radicals even exists in general. However, as long as we are only interested in rational solutions, there exists a complete algorithmic procedure to find them all; this is contained in 1.1.2.

Theorem 1.1.2 (Rational Root). Let $d \in \mathbb{Z}_{\geq 1}$, let $a_0, \dots, a_d \in \mathbb{Z}$, and consider the equation

$$a_0x^d + a_1x^{d-1} + \dots + a_d = 0.$$

Suppose that $x = p/q$ is a rational solution to this equation, where $p, q \in \mathbb{Z}$ are coprime and $q \neq 0$. Then $p \mid a_d$ and $q \mid a_0$. In particular, if $a_0 = 1$ (i.e., the equation is monic), then every rational root is an integer.

Remark 1.1.3. Do you see why this gives a complete (algorithmic) answer (in theory, if we can find all factors of a given number) of finding all rational (and hence integral) roots to a single-variable rational polynomial?

Proof of 1.1.2. Since p/q is a solution to the equation, we have

$$a_0p^d + a_1p^{d-1}q + \dots + a_dq^d = 0.$$

Therefore, $q \mid a_0 p^d$; since $(p, q) = 1$, this forces $q \mid a_0$. The part with p is similar and left to the reader. ■

Exercise 1.1.4. Finish the proof by showing that $p \mid a_d$. You might have to be slightly more careful than for q because p could be zero; what would that mean?

We didn't really use any special properties of the integers here—other than the fact that they admit unique factorization. Therefore, the same result and proof works also with $\mathbb{Z}$ replaced by any unique factorization domain (UFD) R ; see 2.1.2.

Example 1.1.5 (Plane Curves). The next simplest case of $n = 2$ corresponds to *plane curves*. These correspond to a single equation relating two variables $x = x_1$ and $y = x_2$; the corresponding locus can be plotted as a subset of the Euclidean plane $\mathbb{R}^2$.

- (a) When $d = 1$, we are talking about linear equations of the form $ax + by + c = 0$, with $a, b, c \in \mathbb{Q}$ and not both a and b zero. The theory of rational solutions here is familiar from high-school coordinate geometry, and the theory of integer solutions is also well-understood, although it might need a little thought (2.1.3).
- (b) When $d = 2$, we are looking at *conic sections*, so named because these curves can be obtained by slicing a (doubly infinite) cone by planes at various angles (see this excellent video by 3Blue1Brown illustrating this). Examples include circles (e.g., $x^2 + y^2 = 1$), parabolas (e.g., $y = x^2$), ellipses (e.g., $x^2 + 2y^2 = 1$), hyperbolas (e.g., $xy = 1$ or $3x^2 - 2y^2 = 1$), pairs of lines (e.g., $y^2 - x^2 = 0$), and even “doubled lines” (e.g., $x^2 = 0$) (whatever that means). Plot these on Desmos! Here, the theory of rational solutions is very well-understood. These curves may or may not have any rational points (e.g., $x^2 + y^2 = 1$ and $x^2 + 3y^2 = 2$ respectively; see 2.1.4(a)), but if there is at least one, then we understand all of them very well. We will discuss this more next time.
- (c) When $d = 3$, we are looking at *cubic curves*. Again these may or may not have any rational points at all (e.g., $y^2 = x^3 + 1$ and $x^3 + 2y^3 = 4$ respectively; see 2.1.4(b)), but the theory of cubic curves, even those with rational points (these are called *elliptic curves*), is quite subtle. The object of this summer will be to study precisely these objects.

We mention in closing that the cases $d \geq 4$ or $n \geq 3$ are even more complicated; the study of such diophantine equations properly belongs to the field of math known as *arithmetic geometry*. We will not say much about such objects in general, except perhaps for some specific examples.

In 1.1.5, more or less the same discussion can be had with $\mathbb{Q}$ replaced with essentially any field K . (We will systematically make this replacement and work with general fields starting very soon.)

For another Diophantine equation, consider

Example 1.1.6 (Bachet's Example). When does a cube differ from a square by 1? Phrased algebraically, this amounts to solving the equation

$$y^2 = x^3 + 1$$

in pairs of integers (x, y) . Five such solutions are $(-1, 0)$, $(0, \pm 1)$, and $(2, \pm 3)$. Are there more?

More generally, given a $c \in \mathbb{Z}$, we can ask for integer solutions to the equation

$$y^2 = x^3 + c. \tag{1.1.7}$$

The case $c = -2$ was studied extensively by French mathematicians Fermat and Bachet in the early 17th century; see the discussion of the history of this problem in [2, Ch. 17, §10]. Bachet observed in 1621 that if (x_0, y_0) is a solution to (1.1.7) with $y_0 \neq 0$, then so is

$$(x_1, y_1) := \left(\frac{x_0^4 - 8cx_0}{4y_0^2}, \frac{-x_0^6 - 20cx_0^3 + 8c^2}{8y_0^3} \right). \tag{1.1.8}$$

This is quite crazy! It's very easy algebra to verify that it works, but how does one come up with such a thing?

Let's assume this for a moment. Given a “seed solution” (x_0, y_0) , we can iterate the map $(x_0, y_0) \mapsto (x_1, y_1)$ to produce an infinite sequence (x_n, y_n) of solutions to 1.1.7. For instance, if $c = 1$, then starting from the seed $(x_0, y_0) = (0, 1)$ returns the sequence

$$(0, 1), \quad (0, 1), \quad (0, 1), \dots$$

Boring! If we start from the seed $(2, 3)$, then we get the sequence

$$(2, 3), \quad (0, -1), \quad (0, -1), \dots$$

That's slightly more interesting; it's cool that it stabilizes.

Let's take a different c . If we take $c = 3$ and the seed $(x_0, y_0) = (1, 2)$ gets us the sequence

$$(1, 2), \quad \left(\frac{-23}{4^2}, \frac{11}{4^3}\right), \quad \left(\frac{2540833}{88^2}, \frac{4050085583}{88^3}\right), \dots$$

That's very interesting! These numbers seem to get pretty huge pretty quickly. (We will prove later that the number of decimal digits in the numerator and denominator in lowest terms roughly doubles with each iteration.)

When do we get infinitely many distinct solutions in the sequence (x_n, y_n) ? It is a theorem of Fueter from 1930 (reproven by Mordell in 1966), which we will prove later, that if $c \notin \{1, -432\}$, then for any seed (x_0, y_0) with $x_0 y_0 \neq 0$, the sequence (x_n, y_n) gives us infinitely many rational solutions.

What about integer solutions? It is a theorem of Thue from 1908 that for any nonzero c , there are only finitely many integer solutions (x, y) to 1.1.7. Therefore, even if there are infinitely many solutions produced by Bachet's formula (as for $c \notin \{1, -432\}$), only finitely many can be integral. We may or may not get to this result by the end of the course; if you're particularly interested in seeing a proof, I can try to incorporate it.

All of this still begs the question: where does Bachet's formula (1.1.8) really come from? The beautiful answer is that it really comes from *geometry*.¹

The equation $y^2 = x^3 + c$ defines a *cubic curve*, say C , in the plane. This cubic curve has the property that it is *smooth*, i.e., has a well-defined tangent line at every point; equivalently, at every point of the curve, at least one of the partial derivatives $2y$ or $3x^2$ is nonzero.² Given a point $P_0 = (x_0, y_0)$ on this cubic curve C , we can draw the tangent line L to C at P_0 . Then an elementary case of *Bézout's Theorem* tells us that L will intersect C in a third point P_1 on C ; this is quite clear if we parametrize L and restrict the equation of C via this parametrization to get a cubic equation in one variable whose roots correspond to the intersection points. "Two" of these intersection points are at P_0 (i.e., P_0 corresponds to a double root of this cubic equation), and since a cubic has three roots, there must be a third point—this is P_1 . The coordinates of P_1 are exactly the solution (x_1, y_1) expressed in Bachet's formula. In older texts, this procedure $P_0 \mapsto P_1$ is called the "chord and tangent process".

Let's work this out explicitly. Suppose we are given a point (x_0, y_0) on the curve (1.1.7). An easy exercise in elementary geometry or calculus tells us that the slope of the curve 1.1.7 at (x_0, y_0) when $y_0 \neq 0$ is $(3x_0^2)/(2y_0)$, so the equation to the tangent line is

$$y = \left(\frac{3x_0^2}{2y_0}\right)x + \left(y_0 - \frac{3x_0^3}{2y_0}\right). \quad (1.1.10)$$

Substitute this expression for y into 1.1.7 to get the cubic equation

$$x^3 - \left[\left(\frac{3x_0^2}{2y_0}\right)x + \left(y_0 - \frac{3x_0^3}{2y_0}\right)\right]^2 + c = 0.$$

This cubic equation has a double root at $x = x_0$. Let's call its third root x_1 . Then Vieta's formula allow us to compute the the product of the roots as the negative of the constant term, which gives us

$$x_0^2 x_1 = \left(y_0 - \frac{3x_0^3}{2y_0}\right)^2 - c,$$

giving an expression for x_1 in terms of x_0, y_0 , and c . Plugging this into 1.1.10 gives us our y_1 . A little bit of simplification using that (x_0, y_0) lies on 1.1.7—left to the reader (2.1.5)—then expresses (x_1, y_1) in the promised form 1.1.8.

¹I say *really*, because this is probably not how Bachet found it.

²This requires us to work over a field of characteristic other than 2 or 3, so $\mathbb{Q}$ is certainly OK.

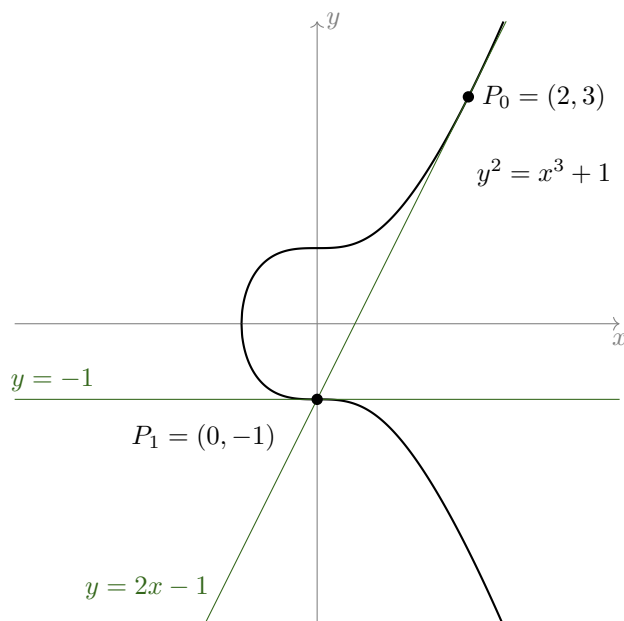


Figure 1.1.9: The “chord and tangent process” which yields Bachet’s formula, illustrated for $c = 1$ and seed $P_0 = (x_0, y_0) = (2, 3)$.

The above procedure also has an analog when you are given two points, say P, Q on C : you take the line L joining P and Q , and consider the third intersection points R of P and Q with C (this is the “chord” part of the “chord and tangent process”). It is a fascinating conjecture of Poincaré from around 1901, proven by Mordell 1922 (and then generalized by Weil to number fields in 1928), which says that given any such smooth cubic C over $\mathbb{Q}$, there are finitely many “seed points” $P_i \in C(\mathbb{Q})$ such that *every* other rational point on C can be obtained by iteratively applying the chord and tangent processes to the points P_i . This is now known as the Mordell Theorem or the Mordell-Weil Theorem, and one of our goals for this summer will be to prove this. This cool interplay between algebra, number theory, and geometry is the tip of the iceberg which is the theory of elliptic curves, and what we shall study this summer.

1.2 26/06/16 - Introduction II: Pythagorean Triples and Fermat for Exponent 3

Let us now study some special cases of the *Fermat problem*, the statement of which is colloquially known as *Fermat's Last Theorem*. Recall that this says that if $n \in \mathbb{Z}_{\geq 3}$ and $X, Y, Z \in \mathbb{Z}$ are such that $X^n + Y^n = Z^n$, then the solution (X, Y, Z) is trivial, i.e., $XYZ = 0$. We will discuss and prove the cases $n = 3$ and $n = 4$.

1.2.A Pythagorean Triples

Before we do that however, it is helpful to review the case $n = 2$ of Pythagorean triples. Here we are trying to solve $X^2 + Y^2 = Z^2$ in the integers. The first thing we note is that this is a *homogeneous equation*; this means that if (X, Y, Z) is a solution, then so is $(\lambda X, \lambda Y, \lambda Z)$ for any λ . This is very helpful, and allows some geometric and arithmetic simplifications. We will develop the language of projective curves in the next lecture to deal with the geometric situation more generally, and we will deal with the arithmetic implications below, but for now it suffices to note that if $Z \neq 0$ (the solutions with $Z = 0$ are easy to understand!), then we can divide throughout by Z to get that $(x, y) := (X/Z, Y/Z)$ satisfies

$$x^2 + y^2 = 1.$$

This curve is familiar to us as the circle C . Thus, Pythagorean triples give rise to points on the circle. (This is a fundamental insight going back even to Descartes!) What is more, if X, Y, Z are integers, then x and y are rational numbers; therefore, if we have a good understanding of rational points on the circle $x^2 + y^2 = 1$, then we will have a way to find all Pythagorean triples. For instance, if we could parametrize all points on the circle, then we can hope to parametrize all Pythagorean triples.

Now the circle is a curve of degree two, i.e., a conic. This has the important consequence that a general line meets it in *two points* (some are tangent, and some might need you to allow complex numbers). This leads us to the following simple strategy: if we pick any rational point P on C and project from it to a general line L , then we will get (essentially) a bijective correspondence between points on this line and on the circle. (Make sure you believe this!) Since it is very easy to parametrize points on a line, this gives us a way of parametrizing points on the conic curve. Let's see this work out in practice.

Take the point P to be $(-1, 0)$ and the line L to be the x -axis. Given a point (x_0, y_0) on the circle C , the line joining P and (x_0, y_0) meets the line L in the point $(0, t)$ where $t = y_0/(x_0 + 1)$. Note that this does not work when $x_0 = -1$ corresponding to P itself, when the limiting line is the tangent line $x = -1$, which does not meet L in the affine plane. We'll figure out how to fix this next time.

Going the other way, we can parametrize points on L as $(0, t)$ for varying t . The line joining $(-1, 0)$ and $(0, t)$ has equation

$$y = tx + t,$$

which we would like to intersect with the circle. Plugging this in to the equation for the circle gives us the quadratic equation

$$x^2 + (tx + t)^2 = 1,$$

which factors as

$$(x + 1)((t^2 + 1)x + (t^2 - 1)) = 0.$$

One root is the expected $x = -1$; it's the other one which we are after, which has $x = (1 - t^2)/(1 + t^2)$. Using $y = tx + t$, this gives us the other point of intersection

$$(x, y) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right).$$

(Those who have seen a little trigonometry might recognize this is as the formula for $(\cos \theta, \sin \theta)$ in terms of $\tan(\theta/2)$; why is that?). These operations are visibly inverses to each other and the resulting maps

$$(x, y) \mapsto y/(x + 1) \text{ and } t \mapsto \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right)$$

are almost inverses to each other (check!). The only point that gives us some trouble is the point $P = (-1, 0)$ itself, which should somehow correspond to “ $t = \infty$ ”. Now, the above discussion works over any field (but see 1.2.1). In particular, if t is a rational number, then so is (x, y) , and conversely. In this way, we have completely figured out all rational points on C .

Remark 1.2.1. So far we haven’t really used many properties of the rational numbers. The same algebra can be carried out over more or less any field K (even if it’s not easy to visualize what that might mean), with some important caveats. Here are two things to think about:

- (a) What (if anything) goes wrong when you work with a field of characteristic two such as $K = \mathbb{F}_2$ instead?
- (b) If we work over the field $K = \mathbb{C}$, what happens to the values $t = \pm i$ of the parameter t ?

Remark 1.2.2. Note that we could have “dehomogenized with respect to other variables” too. For instance, if $Y \neq 0$, then we may set $(s, t) := (X/Y, Z/Y)$ instead to get instead the equation $s^2 + 1 = t^2$, or equivalently $t^2 - s^2 = 1$. This would have led to a parametrization similar to the one above. This additional symmetry—while not useful here—can be extremely useful in other contexts; we will study this systematically next time.

Let us see what consequence this has for the arithmetic. Suppose we are interested in finding integer solutions X, Y, Z to $X^2 + Y^2 = Z^2$ with $X, Y, Z \neq 0$. We may assume by flipping signs if needed that $X, Y, Z > 0$. We may also divide throughout by the greatest common divisor of all X, Y, Z to assume that X, Y, Z have no common factors as a *triple*. This has the consequence that X, Y, Z have no common factors *pairwise*: if a prime p divides two of X, Y, Z , then the equation $X^2 + Y^2 = Z^2$ forces it to divide the third. (Therefore, things like $(6, 10, 15)$ will never work.) With these simplifications, we can show

Theorem 1.2.3. Let X, Y, Z be pairwise coprime positive integers such that $X^2 + Y^2 = Z^2$. Then there are unique coprime $m, n \in \mathbb{Z}$ of different parity with $m > n > 0$ such that exactly one of (X, Y, Z) or (Y, X, Z) is

$$(m^2 - n^2, 2mn, m^2 + n^2).$$

Proof. The point $(x, y) = (X/Z, Y/Z)$ lies on the circle $C : x^2 + y^2 = 1$ and is different from $(-1, 0)$, so the above discussion tells us that there is a unique $t \in \mathbb{Q}$ such that

$$\left(\frac{X}{Z}, \frac{Y}{Z} \right) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right).$$

Since x and y are positive, we conclude that $0 < t < 1$ (check!). Write $t = n/m$ for coprime positive $m, n \in \mathbb{Z}$ such that $m > n > 0$, so that

$$\left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2} \right) = \left(\frac{m^2 - n^2}{m^2 + n^2}, \frac{2mn}{m^2 + n^2} \right).$$

We claim that the fractions on the right are in their lowest terms (up to small factors of two; see below). Why is this?

Let’s analyze the parities of m and n . Since they are coprime, they can’t both be even. If they have different parity, then $m^2 \pm n^2$ are odd, and this forces

$$\gcd(m^2 + n^2, m^2 - n^2) = \gcd(2mn, m^2 + n^2) = 1.$$

i.e., these fractions are in lowest form (check!). This proves that (X, Y, Z) is of the form desired.

Finally, can they both be odd? If this is the case, then the integers $m_1 = (m + n)/2$ and $n_1 := (m - n)/2$ are coprime, satisfy $m_1 > n_1 > 0$, have different parity (check!), and further we have that

$$\left(\frac{m^2 - n^2}{m^2 + n^2}, \frac{2mn}{m^2 + n^2} \right) = \left(\frac{2m_1n_1}{m_1^2 + n_1^2}, \frac{m_1^2 - n_1^2}{m_1^2 + n_1^2} \right),$$

and so the argument from the previous paragraph implies $(Y, X, Z) = (m_1^2 - n_1^2, 2m_1n_1, m_1^2 + n_1^2)$, finishing the proof. ■

Exercise 1.2.4. Fill in the details in the above proof: show that if $m, n \in \mathbb{Z}$ are coprime and of different parity, then $\gcd(m^2 + n^2, m^2 - n^2) = \gcd(2mn, m^2 + n^2) = 1$, and show that if $m > n > 0$ are both odd, then $m_1 = (m + n)/2$ and $n_1 := (m - n)/2$ are coprime and have different parity.

In the above study, we didn't really use any properties of the curve C except that it is of degree two—and that it had a rational point P to get our method started. Therefore, the same argument applies to any conic C with a rational point $P \in C(\mathbb{Q})$. This allows us to give complete parametrizations of all rational points of such objects; see, for instance, 2.1.6(a). (Get a hold on the integer points this way might be a little harder; see 2.1.6(b).)

Clearly, then we would have a complete understanding of rational solutions to all conic sections C if we could just prove that they all contained some rational point P . Sadly, this is not the case (2.1.4(a)). How can we figure out when a given conic has a rational point, and when it doesn't?

Example 1.2.5. Let's look at a few examples.

(a) Does the curve

$$4x^2 + 12xy + 9y^2 + 1 = 0$$

have any rational points? No. This is because it can be written as $(2x + 3y)^2 + 1 = 0$, which doesn't have any real solutions (x, y) , let alone rational ones. We say such curves are *locally obstructed at ∞* or *locally obstructed over $\mathbb{R}$* .

(b) Does the curve

$$x^2 + xy + y^2 = 2$$

have any rational points? No, but this is not because it doesn't have any real points (e.g., $(\sqrt{2}, 0)$ works). This is more subtle. Suppose there is a rational solution (x, y) . Write $x := X/Z$ and $y := Y/Z'$ for integers $X, Y, Z, Z' \in \mathbb{Z}$ with $Z, Z' > 0$ and $\gcd(X, Z) = \gcd(Y, Z') = 1$. Plugging this into the equation and multiplying through by $Z^2(Z')^2$ gives us

$$X^2(Z')^2 + XYZZ' + Y^2Z^2 = 2Z^2(Z')^2.$$

This equation implies that $Z \mid (Z')^2$ and so by symmetry that $Z' \mid Z^2$. This is enough to conclude that Z and Z' have the same prime factors, but this is not enough to conclude that $Z = Z'$ (consider 2^23^3 vs. 2^33^2). We need to be a little more careful. For this, let p be a prime dividing both Z and Z' , and let $\alpha := v_p(Z)$ (resp. $\alpha' := v_p(Z')$), so $\alpha, \alpha' \geq 1$. We have that $v_p(X) = v_p(Y) = 0$, that $\alpha \leq 2\alpha'$ and $\alpha' \leq 2\alpha$. We have to show that $\alpha = \alpha'$. Suppose to the contrary that $\alpha \neq \alpha'$; we may then assume without loss of generality that $\alpha' > \alpha$ (the other case being symmetrical). In this case, the p -adic valuation of the terms on the left is $2\alpha', \alpha + \alpha'$, and 2α , which forces the sum to have p -adic valuation 2α . The right hand side has v_p equal to either $2\alpha + 2\alpha$ if $p \neq 2$, and $2\alpha + 2\alpha' + 1$ if $p = 2$, and we easily see that none of these options work. Thus, we must have $\alpha = \alpha'$. Since this is true for all p and $Z, Z' > 0$, we conclude that $Z' = Z$, and so we can remove it to get the homogeneous equation

$$X^2 + XY + Y^2 = 2Z^2 \tag{1.2.6}$$

we were hoping for. Now we are almost done. Let's consider the parities of X and Y . Well, it cannot be that both of them are odd or that precisely one is odd, because then the left side of 1.2.6 is odd. Therefore, both of them must be even, say $X = 2X_1$ and $Y = 2Y_1$. This then gives us the equation $2(X_1^2 + X_1Y_1 + Y_1^2) = Z^2$, which forces Z to be even as well—but $\gcd(X, Z) = 1$, and this is a contradiction. This proves that there couldn't have been any solutions to begin with.

We say that the curve $x^2 + xy + y^2 = 2$ is *locally obstructed at 2*. Note that this does not mean it has no solutions mod 2 (i.e., that $C(\mathbb{F}_2) = \emptyset$, which is false); rather, it means that it has no *2-adic solutions*, i.e., $C(\mathbb{Q}_2) = \emptyset$. (Don't worry if you don't know yet what that means.)

Similarly, curves can be locally obstructed at other primes; see 2.1.7.

For right now, you can take local obstruction at a prime $p > 0$ to mean that “the arithmetic doesn't work out at p ”. Clearly, if there is a local obstruction at any prime (including the “prime” $p = \infty$ corresponding to $\mathbb{Q}_\infty = \mathbb{R}$), then there is no rational solution. But the converse is not that clear. Can we have a curve that has solutions *everywhere locally*, but none globally?

It turns out that this “failure of the local-to-global principle” cannot happen for conics, but it can happen for cubics. The fact that it cannot happen to conics is known as the Hasse-Minkowski

Theorem. This holds more generally not just for conic curves, but for quadratic forms in any number of variables. Proving this would take us a little far afield, so I will leave the simplest case of quadratic equations as an exercise (2.1.10), and refer the reader to [3, Ch. IV, Thm. 8] for the general proof.

The really interesting thing I want to draw your attention to, however, is that this result fails in general for curves of degree ≥ 3 : there are cubic curves which have solutions in $\mathbb{Q}_p$ for all $p \leq \infty$, but no solutions in $\mathbb{Q}$. One of the simplest examples of this sort is the Selmer curve

$$3X^3 + 4Y^3 + 5Z^3 = 0.$$

(This is the homogeneous equation; the inhomogeneous equation is $3x^3 + 4y^3 = -5$.) We will revisit this curve later in the course, and develop tools to study such “obstructions to the Hasse-Minkowski principle”. For now, let us turn to

1.2.B Fermat for Exponent 3

The next case of Fermat’s problem is $n = 3$, where we are solving $X^3 + Y^3 = Z^3$. Again, this is a homogeneous equation. It is now believed that Fermat did not have a proof for this case, and that the first proof was given by Euler in 1770 (although see the discussion on the Wikipedia page). Euler’s proof used the arithmetic of $\mathbb{Z}[\omega]$, where $\omega = (-1 + \sqrt{-3})/2$ is a primitive cube root of unity.

We’ll pursue a totally different idea. The idea is that the curve $X^3 + Y^3 = Z^3$ is a cubic plane curve (again, given the understanding of (de)homogenization), and has a rational inflection point; this allows us to put it in *Weierstrass form*. More generally, we can consider for any nonzero α the equation

$$X^3 + Y^3 = \alpha Z^3.$$

This equation has rational point $[1 : -1 : 0]$. Let’s move this a little bit, specifically to $[0 : 1 : 0]$: consider the “change of coordinates” given by

$$(X_1, Y_1, Z_1) = (12\alpha Z, 36\alpha(X - Y), X + Y). \quad (1.2.7)$$

Exercise 1.2.8. Figure out how to invert this change of coordinates: express (X, Y, Z) in terms of (X_1, Y_1, Z_1) .

We will see below why this change of coordinates is a good one; we will explain later where it comes from. Given this “inverse change of coordinates”, we note that the curve above transforms to the one of the form

$$Y_1^2 Z_1 = X_1^3 - 432\alpha^2 Z_1^3,$$

which after dehomogenizing (dividing throughout by Z_1^3) is the curve

$$y^2 = x^3 - 432\alpha^2.$$

This is nothing but our example from 1.1.6, with $c = -432\alpha^2$, and we are back to finding rational points on this cubic curve. The Fermat case corresponds to $\alpha = 1$, and hence $c = -432$. (This might explain why $c = -432$ is special in 1.1.6.) This is again what we call an *elliptic curve*. We will show later using techniques for elliptic curves that the only points on here are the points $(12, \pm 36)$.

Exercise 1.2.9. Which points on $X^3 + Y^3 = Z^3$ do the points $(12, \pm 36)$ on $y^2 = x^3 - 432$ correspond to?

This finishes the proof of Fermat’s last theorem modulo the claim made about the curve $y^2 = x^3 - 432$, which we will prove later. I also still need to explain where this change of coordinates comes from; that will be done sooner. (Hopefully this motivates you to keep coming to this course!) We’ll deal with the case $n = 4$ next time.

1.3 26/06/17 - Introduction III: Fermat for Exponent 4 and Infinite Descent; Review of Projective Geometry I: Basics

Let's first finish up our discussion of the Fermat problem for the exponent $n = 4$.

1.3.A Fermat for Exponent 4, Proof 1

Let's try to solve the equation

$$X^4 + Y^4 = Z^4.$$

This is the only case for which Fermat is believed to have a proof, and Fermat's proof uses a very important strategy, known now as "Fermat's method of infinite descent". Let us now review (in modern terminology) Fermat's proof. In fact, we will show something stronger.

Theorem 1.3.1 (Fermat). Let $X, Y, W \in \mathbb{Z}_{\geq 0}$ be such that $X^4 + Y^4 = W^2$. Then $XYW = 0$.

Proof. Suppose to the contrary there is a solution (X, Y, W) with $X, Y, W > 0$, and pick such a solution with the smallest possible W . This implies in particular that X^2, Y^2, W are pairwise coprime (why?) and hence using 1.2.3 that (possibly up to permuting X and Y , which we do), there are unique coprime $m, n \in \mathbb{Z}$ of different parity with $m > n > 0$ such that

$$(X^2, Y^2, W) = (m^2 - n^2, 2mn, m^2 + n^2);$$

in particular, X is odd. Now $X^2 = m^2 - n^2$, so that $X^2 + n^2 = m^2$. Again X, n, m are pairwise coprime positive integers with X odd, so 1.2.3 applied once again gives us unique coprime $p, q \in \mathbb{Z}$ of different parity with $p > q > 0$ and such that

$$(X, n, m) = (p^2 - q^2, 2pq, p^2 + q^2).$$

In particular, n is even and hence m is odd. Finally, $m = p^2 + q^2$ implies that m, p , and q are pairwise coprime. Therefore

$$Y^2 = 2mn = 4mpq$$

implies that p, q, m are all squares. Write $(p, q, m) = (X_1^2, Y_1^2, W_1^2)$ for some integers $X_1, Y_1, W_1 > 0$. Then

$$X_1^4 + Y_1^4 = p^2 + q^2 = m = W_1^2,$$

so that (X_1, Y_1, W_1) is also a solution. Finally,

$$W_1 \leq W_1^2 = m \leq m^2 < m^2 + n^2 = W$$

tells us that this is a "smaller" solution, giving us the required contradiction. ■

Can you see why this is called "infinite descent"?

The machinery of elliptic curves (specifically descent on them) is a vast modern generalization of this technique due to Fermat; this is where the name "descent" in the theory of elliptic curves comes from. Now Fermat's equation for $n = 4$ does not lend itself directly to the technology of elliptic curves, because it not one: $X^4 + Y^4 = Z^4$ defines a *quartic* curve in the plane which is a *canonical curve of genus two* (whatever that means). However, there is a clever way to massage it to give points on an elliptic curve; it is this curve that we do "descent" on. Since it's fun to figure out how to do this massaging, I will leave this as an exercise (2.1.12). We will figure out how to do descent on that curve—and hence prove Fermat's result in a second way—in a more general context much later.

In closing, let me remark that the case $n \geq 5$ of Fermat's theorem is genuinely very difficult, and it is beyond my abilities to go into the details here (see the history on the Wikipedia page, for instance). However, Andrew Wiles's proof from 1994 of it still nontrivially uses the theory of elliptic curves, following an idea going back to Frey from 1984. If we have time and interest, I am happy to explain the outline of the proof towards the end of the course; let me know if this is something you would like to see.

To fruitfully study plane curves, we need some basic understanding of the projective geometry of the plane. Let's take a detour to rapidly review what we need; a much fuller account can be found at [4].

1.3.B Basics

Example 1.3.2. Fix a field K , and for $s \in K$ consider the intersection of the two lines

$$sy = x \text{ and } x = 1.$$

When $s \neq 0$, there is a unique intersection point, namely $(1, 1/s)$, but when $s = 0$, there is no intersection at all, and the lines are parallel. What has happened is that as $s \rightarrow 0$, the intersection point has “escaped to infinity”.

Such behavior can be extremely troublesome while doing geometry, and it is helpful to develop basic vocabulary to deal with “points at infinity”. The modern approach to doing this involves working with *homogeneous coordinates on projective space*.

Definition 1.3.3. Let K be a field and $n \in \mathbb{Z}_{\geq 0}$. We define the *set of K -points of projective n -space over K* to be

$$\mathbb{P}^n(K) := \{[X_0 : X_1 : \cdots : X_n] : X_0, \dots, X_n \in K \text{ not all zero}\} / \sim$$

where $\sim$ means that we identify points with the same coordinates up to scaling by a nonzero number, i.e., for all $X_0, \dots, X_n \in K$, not all zero, and $\lambda \in K^\times$ we have

$$[X_0 : \cdots : X_n] \sim [\lambda X_0 : \cdots : \lambda X_n].$$

The coordinates $X_0, \dots, X_n$ are called *homogeneous coordinates*, and are well-defined only up to simultaneous scaling.

Unimportant Remark 1.3.4. For those who have seen linear algebra, the set $\mathbb{P}^n(K)$ can be identified with the set of one-dimensional K -linear subspaces of K^{n+1} (how?). For those who have seen a little topology, when $K = \mathbb{R}$ or $K = \mathbb{C}$ (or more generally any topological field such as $K = \mathbb{Q}_p$), then $\mathbb{P}^n(K)$ can be given a natural topology, which for $K = \mathbb{R}, \mathbb{C}, \mathbb{Q}_p$ even makes it a(n) (analytic) manifold.

Example 1.3.5.

- (a) When $n = 0$, the space $\mathbb{P}^0(K)$ is just one point.
- (b) When $n = 1$, the points of $\mathbb{P}^1(K)$ look like $[S : T]$, where $S, T \in K$ are not both zero, and we can simultaneously scale S and T by the same nonzero element λ of K . In particular, when $S \neq 0$, then scaling by $\lambda = S^{-1}$ gives the unique representative $[1 : t]$ where $t := T/S \in K$. In this way, $\mathbb{A}^1(K)$ sits naturally in $\mathbb{P}^1(K)$. The only point not contained in it is the point $[0 : 1]$ corresponding to $S = 0$; this can be thought of as corresponding to “ $t = 1/0 = \infty$ ” and will often be called the *point at infinity* and denoted ∞ . Therefore, $\mathbb{P}^1(K) = \mathbb{A}^1(K) \amalg \{\infty\}$ is the “one-point compactification” of $\mathbb{A}^1(K)$. (Warning: This is not true for higher n .) This allows us to handle infinity easily—no actual division by zero needed.
- (c) When $n = 2$, we are looking at triples $[X : Y : Z]$ of $X, Y, Z \in K$, not all zero, up to simultaneous scaling. Again, if $Z \neq 0$, then scaling by $\lambda = Z^{-1}$ gives us the unique representative $[x : y : 1]$ where $(x, y) = (X/Z, Y/Z)$, and again we get an $\mathbb{A}^2(K) \subset \mathbb{P}^2(K)$. The complement corresponds to $Z = 0$, which is the set of points $[X : Y : 0]$, in natural bijection with $\mathbb{P}^1(K)$ itself. This locus is called the *line at infinity*; the point $[X : Y : 0]$ should be thought of as the point infinitely far away in the direction of the vector $[X : Y]$. In this way, $\mathbb{P}^2(K) = \mathbb{A}^2(K) \amalg \mathbb{P}^1(K)$. Said yet another way, we have

$$\{[X : Y : Z] \in \mathbb{P}^2(K) : Z \neq 0\} = \{[X/Z : Y/Z : 1] \in \mathbb{P}^2(K)\} = \{(x, y) \in \mathbb{A}^2(K)\},$$

where again $(x, y) = (X/Z, Y/Z)$.

Unimportant Remark 1.3.6.

(a) Continuing in the above fashion, we can describe a decomposition of $\mathbb{P}^n(K)$ of the form

$$\mathbb{P}^n(K) = \mathbb{A}^n(K) \amalg \mathbb{P}^{n-1}(K) = \mathbb{A}^n(K) \amalg \mathbb{A}^{n-1}(K) \amalg \cdots \amalg \mathbb{A}^1(K) \amalg \mathbb{A}^0(K).$$

- (b) We will probably not seriously need $\mathbb{P}^n(K)$ for $n \geq 3$ in this course, but it may be helpful to note that much of what follows will apply with minor modifications to the general case.
- (c) If you have seen barycentric coordinates before, then homogeneous coordinates on projective space $\mathbb{P}^2$ can be thought of as barycentric coordinates with the coordinate (or pure weight) points at $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$.

All faults of affine geometry are rectified in projective geometry, and this boils down to basic linear algebra/matrix theory; as an example, let's see how to fix 1.3.2.

Example 1.3.7. The “projectivization” of the lines of 1.3.2 are the lines

$$sY = X \text{ and } X = Z$$

in $\mathbb{P}^2(K)$. These two *always* have a unique intersection point, namely $[s : 1 : s]$, in $\mathbb{P}^2(K)$. Note that for $s = 0$, this is the point $[0 : 1 : 0]$ at infinity along the vertical direction, as expected.

Example 1.3.8. The “projectivization” of the parametrization in §1.2.A is the map $\mathbb{P}^1 \rightarrow \mathbb{P}^2$ given by

$$[S : T] \mapsto [S^2 - T^2 : 2ST : S^2 + T^2],$$

with image $X^2 + Y^2 = Z^2$. (Why is this map well-defined? For this to be true, we have to check that $S^2 - T^2, 2ST, S^2 + T^2$ are never simultaneously zero. This requires our base field to have characteristic other than 2.) When $\text{ch } K \neq 2$, this gives us a *genuine* bijection (even isomorphism) between $\mathbb{P}^1(K)$ and $C(K)$, where $C = \mathbb{V}(X^2 + Y^2 - Z^2) \subset \mathbb{P}_K^2$. This way of looking at it solves various mysteries: for instance, the unique point mapping to $P = (-1, 0) = [-1 : 0 : 1]$ is precisely $[S : T] = [0 : 1]$, also known as “ $t = \infty$ ”. Similarly, over $K = \mathbb{C}$ the points $[1 : \pm i]$ map to the points $[1 : \pm i : 0]$ in $\mathbb{P}^2(\mathbb{C})$, known as the “circular points at infinity”; these are not visible over $\mathbb{Q}$ or $\mathbb{R}$.

Let us study solutions of polynomial equations on projective spaces; for simplicity, we will stick to $n = 3$ and use homogeneous coordinates X, Y, Z . On projective space, the vanishing locus of an arbitrary $F \in k[X, Y, Z]$ does not make sense in general: if $F = X^2 - Y$, for instance, then $(1, 1, 0)$ lies on F but $(2, 2, 0)$ does not. However, the one case in which this problem does *not* arise is when F is *homogeneous*, i.e., when all monomials appearing in F have the same degree, say $d \in \mathbb{Z}_{\geq 0}$ (c.f. 2.1.11). In this case we have

$$F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z)$$

for $\lambda \in K^\times$, so it makes sense to speak of the *curve* $C = \mathbb{V}(F)$ cut out by F in projective space. In other words, C consists of K -points

$$C(K) := \{[X : Y : Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0\}.$$

Example 1.3.9. If $F = Y^2Z - X^3 - Z^3$ and $C = \mathbb{V}(F) \subset \mathbb{P}_K^2$, then $\mathcal{O} := [0 : 1 : 0] \in C(K)$ but $[1 : 0 : 0] \notin C(K)$.

Clearly, replacing F by a nonzero scalar multiple μF for $\mu \in K^\times$ is not going to change the vanishing locus, and in this way we get an identification between curves and nonzero homogeneous polynomials in $K[X, Y, Z]$ up to scaling. We take this to be our definition.

Definition 1.3.10. A *projective plane curve* C over a field K is an equivalence class of nonzero homogeneous polynomials $F \in K[X, Y, Z]$ under the equivalence relation of scaling: F and μF for $\mu \in K^\times$ define the same curve $C = \mathbb{V}(F) = \mathbb{V}(\mu F)$. A representative F of this equivalence class is said to be a *defining equation* of the curve. The degree of any defining equation is then called the *degree* of the curve.

We emphasize that this is an *algebraic* definition; the curve is determined by its equation and not its set of K -points, of which there may not be any (e.g., $C = \mathbb{V}(X^2 + Y^2 + Z^2)$ over $K = \mathbb{R}$). A curve

of degree one (resp. two, three, four, etc.) is called a *line* (resp. *conic*, *cubic*, *quartic*, etc.). However, it is still natural to think of the curve C —to a first approximation—in terms of its K -points $C(K)$.

How do we visualize a projective curve C ? Let us now briefly discuss something we have seen in practice already. Given a projective curve $C = \mathbb{V}(F) \subset \mathbb{P}_K^2$ over some field K and of degree $d \in \mathbb{Z}_{\geq 1}$, we can dehomogenize it with respect to any of the variables X, Y , or Z to get an affine curve, where dehomogenization, say, with respect to the variable Z amounts to either dividing throughout by Z^d and using coordinates $x := X/Z$ and $y := Y/Z$, or equivalently looking at points in the affine patch $Z \neq 0$, where we can normalize to $Z = 1$. Algebraically, this amounts to replacing $F \in K[X, Y, Z]$ by $f(x, y) := F(x, y, 1) \in K[x, y]$. In other words,

$$\begin{aligned} C(K) \cap \{Z \neq 0\} &= \{[X : Y : Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0 \text{ and } Z \neq 0\} \\ &= \{(x, y) \in \mathbb{A}^2(K) : F(x, y, 1) = 0\} \\ &= \{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}. \end{aligned}$$

There is nothing special about Z here, and we can equally well dehomogenize with respect to say Y , by taking coordinates $t := X/Y$ and $s := Z/Y$ instead. These different “local patches” of the curve can then be glued together to get a fuller picture of the projective curve.

Example 1.3.11. Consider the projective curve $C := \mathbb{V}(Y^2Z - X^3 - Z^3) \subset \mathbb{P}_K^2$ (for any field K). The affine patch $Z \neq 0$ looks like the familiar curve $y^2 = x^3 + 1$, and the affine patch $Y \neq 0$ looks like $s = t^3 + s^3$. The first patch has all points on C except for $\mathcal{O} = [0 : 1 : 0]$; the second patch has all points on C except for $[-\omega^j : 0 : 1]$ for $j = 0, 1, 2$, where $\omega = \zeta_3$ is a primitive cube root of unity. (Some of these point may not exist in $C(K)$. If K has characteristic three, there is only one point $[-1 : 0 : 1]$.) These two patches are related by the coordinate change

$$(t, s) = \left(\frac{x}{y}, \frac{1}{y} \right) \text{ and } (x, y) = \left(\frac{t}{s}, \frac{1}{s} \right).$$

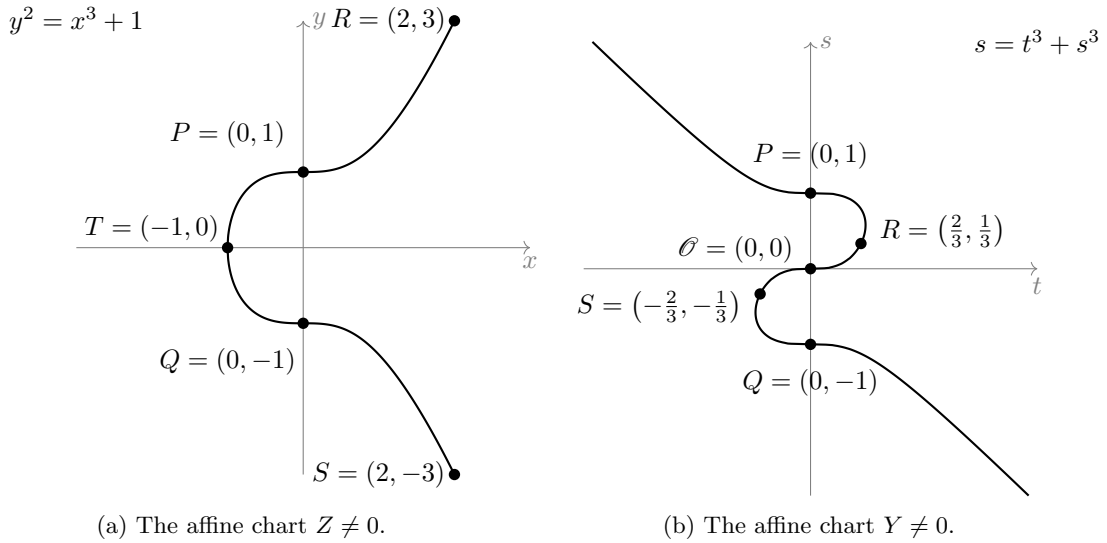


Figure 1.3.12: Two affine patches of the curve $Y^2Z = X^3 + Z^3$ with a few marked points. Note that the point $\mathcal{O}$ is not visible in the first patch, and the point T is not visible in the second patch, but together they cover everything. Note also that the line at infinity in the first patch becomes the t -axis $s = 0$ in the second patch, and that this line has a “triple” intersection with the curve at $\mathcal{O}$, i.e., is an inflectional tangent to C at $\mathcal{O}$.

I would recommend playing with Desmos3D and plotting $Y^2Z = X^3 + Z^3$ and seeing how it intersects the loci $Z = 1$, $Y = 1$, and $X^2 + Y^2 + Z^2 = 1$ to see how these pictures are related to each other.

Going in the other direction, given an affine curve $C \subset \mathbb{A}_K^2$ given by the vanishing of say $f(x, y) \in K[x, y]$, we can look at its *projective closure*. This corresponds to replacing f by its *homogenization* $F \in K[X, Y, Z]$ obtained by replacing x by X/Z and y by Y/Z and clearing denominators, or

(equivalently) by replacing x by X and y by Y , and inserting suitable powers of Z in the monomials to get a homogeneous polynomial of lowest possible degree. This projective closure is usually denoted by $\overline{C} \subset \mathbb{P}_K^2$. We can recover C from $\overline{C}$ by dehomogenizing with respect to Z once again.

Example 1.3.13. The line $2x - 5y + 3 = 0$ in $\mathbb{A}_K^2$ has projective closure $2X - 5Y + 3Z = 0$ in $\mathbb{P}_K^2$. To obtain the points at infinity on this projective closure, we set $Z = 0$ to get the unique point $[5 : 2 : 0]$ in the direction of this line. Make sure this makes sense to you!

Example 1.3.14. The rectangular hyperbola $x^2 - y^2 = 1$ in $\mathbb{A}_K^2$ has projective closure $X^2 - Y^2 = Z^2$ in $\mathbb{P}_K^2$. To obtain the points at infinity of this projective closure, we set $Z = 0$ to get the two points $[1 : \pm 1 : 0]$. Of course, this is as expected: these correspond to the asymptotes of the hyperbola.

Exercise 1.3.15. Are the operations of homogenization and dehomogenization inverses?

1.4 25/06/18: Review of Projective Geometry II: Projective Changes of Coordinates

Perhaps I didn't do a great job of explaining projective space last time, so let me try again.

1.4.A More on Curves and Homogenization

Let us review some things from last time. Given a field K , we met the *affine plane* $\mathbb{A}^2(K)$ over K , which is just the good old plane: it consists of all points (x, y) where $x, y \in K$:

$$\mathbb{A}^2(K) = \{(x, y) : x, y \in K\}.$$

Note that for any field extension L/K , we have a natural inclusion $\mathbb{A}^2(K) \subset \mathbb{A}^2(L)$.

Given a nonzero $f \in K[x, y]$, we can look at its *vanishing locus* in $\mathbb{A}^2(K)$, which is just

$$\{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}.$$

This is what we want to call a curve. Note that for any $\lambda \in K^\times$, the polynomials f and λf have the same vanishing loci; they define the same *curve*. This motivates the algebraic

Definition 1.4.1.

- An *affine curve* is an equivalence class of nonzero $f \in K[x, y]$ under the equivalence relation $f \sim \lambda f$ for nonzero $f \in K[x, y]$ and $\lambda \in K^\times$.

The curve defined by a nonzero f will be denoted $C := \mathbb{V}(f)$, so that $\mathbb{V}(f) = \mathbb{V}(\lambda f)$ for all $\lambda \in K^\times$.

- Such an f as above is called a *defining equation* for the curve C , and the *degree of the curve* C is defined to be the degree of any defining equation for C .

A(n) (affine) curve of degree one (resp. two, three, four, etc.) is called a(n) (affine) *line* (resp. *conic*, *cubic*, *quartic*, etc.).

Given such an f and the associated curve $C = \mathbb{V}(f) \subset \mathbb{A}_K^2$, we can then look at its K -points

$$C(K) := \{(x, y) \in \mathbb{A}^2(K) : f(x, y) = 0\}.$$

The reason for keeping this notation separate is because given a polynomial $f \in K[x, y]$, we would like to study its solutions over any field extension L/K . For instance, K could be $\mathbb{Q}$, and L could be $\mathbb{R}$ or $\mathbb{C}$ or $\mathbb{Q}_p$ or some number field like $\mathbb{Q}[i]$ or $\mathbb{Q}[\sqrt{5}]$. The above notation easily lets us do that, so we get a natural inclusion $\mathbb{A}^2(K) \subset \mathbb{A}^2(L)$ and $C(K) \subset C(L)$.

Example 1.4.2. Let $K = \mathbb{Q}$, let $f(x, y) = x^2 + y^2 - 1 \in \mathbb{Q}[x, y]$, and let $C := \mathbb{V}(f)$. Then $C(\mathbb{Q}) \subset C(\mathbb{R}) \subset C(\mathbb{C})$. We have that $(1, 0) \in C(\mathbb{Q})$, that $(1/\sqrt{2}, 1/\sqrt{2}) \in C(\mathbb{R}) \setminus C(\mathbb{Q})$, and $(i, \sqrt{2}) \in C(\mathbb{C}) \setminus C(\mathbb{R})$.

This algebraic definition also allows us to work with curves that have no points over the base field.

Example 1.4.3. Let $K = \mathbb{Q}$, let $f(x, y) := x^2 + y^2 + 1 \in \mathbb{Q}[x, y]$, and let $C := \mathbb{V}(f)$. Then $C(\mathbb{Q}) = C(\mathbb{R}) = \emptyset$ but $C(\mathbb{C}) \neq \emptyset$, because for instance $(0, i) \in C(\mathbb{C})$. In particular, C is not “empty”; it's just that $\mathbb{R}$ is not a good enough field to see any points on it.

Exercise 1.4.4. Let K be an algebraically closed field. Then for any nonzero $f \in K[x, y]$ of positive degree, if we set $C := \mathbb{V}(f)$, then $C(K) \subset \mathbb{A}^2(K)$ is nonempty. (Note that of course this is not true if K is not algebraically closed, as we have seen above in 1.4.3.)

The abstract curve $C = \mathbb{V}(f) \subset \mathbb{A}_K^2$ arising from this algebraic definition encodes this data of points in every field extension L/K .

Unimportant Remark 1.4.5. The objects $\mathbb{A}_K^2$ and $\mathbb{A}^2(K)$ are slightly different; the latter is the set of K -points of the former (and is probably more properly written $\mathbb{A}_K^2(K)$). The first is a “formal object” that records the data of all $\mathbb{A}^2(L)$ for all extensions L/K . The same goes for the curve C and its L -points $C(L)$ for any L/K . The subscript K in the formal object reminds us what field we are doing the algebra; for instance, if $K = \mathbb{F}_3$ and $C = \mathbb{V}((1 \bmod 3)y + (2 \bmod 3)x + (1 \bmod 3)) \subset \mathbb{A}_{\mathbb{F}_3}^2$, then the subscript reminds us that it makes no sense to talk about $C(\mathbb{Q})$, although it makes perfect sense to talk about $C(\mathbb{F}_3)$ or more generally $C(L)$ for any $L/\mathbb{F}_3$. (For the cognoscenti: $\mathbb{A}_K^2 := \text{Spec } K[x, y]$ and $C := \text{Spec } K[x, y]/(f)$.)

Remark 1.4.6. One strange consequence of the above definition is that sometimes you can have two curves $C, C' \subset \mathbb{A}_K^2$ such that for all fields L/K we have $C(L) = C'(L)$ even though $C \neq C'$. For instance, C and C' might even have different degrees, e.g., if $C = \mathbb{V}(x)$ and $C' = \mathbb{V}(x^2)$. Why do you think this is a good thing? Is it?

More or less the same things as above can be done in the projective plane. As above,

$$\mathbb{P}^2(K) = \{[X : Y : Z] : X, Y, Z \in K \text{ not all zero}\} / \sim$$

where the $\sim$ means that we identify points $[X : Y : Z]$ and $[\lambda X : \lambda Y : \lambda Z]$ for all $\lambda \in K^\times$. As before, for each L/K , we have $\mathbb{P}^2(K) \subset \mathbb{P}^2(L)$. Finally, for any K , the affine space $\mathbb{A}^2(K)$ sits inside $\mathbb{P}^2(K)$ by sending (x, y) to $[x : y : 1]$, with the complement being the “line at infinity” given by $Z = 0$.

Given a nonzero *homogeneous* $F \in K[X, Y, Z]$, we can look at its *vanishing locus* in $\mathbb{P}^2(K)$, which is just

$$\{[X : Y : Z] \in \mathbb{P}^2(K) : F(X, Y, Z) = 0\}.$$

As before, we have

Definition 1.4.7.

- A *projective curve* is an equivalence class of nonzero homogeneous $F \in K[X, Y, Z]$ under the equivalence relation $F \sim \lambda F$ for nonzero homogeneous $F \in K[X, Y, Z]$ and $\lambda \in K^\times$.

The curve defined by a nonzero F will be denoted $D := \mathbb{V}(F)$, so that $\mathbb{V}(F) = \mathbb{V}(\lambda F)$ for all $\lambda \in K^\times$.

- Such an F as above is called a *defining equation* for the curve D , and the *degree of the curve* D is defined to be the degree of any defining equation for D .^a

^aFor the cognoscenti: D stands for “divisor”.

As before, we have projective lines, conics, cubics, quartics, etc., and as before for any $D = \mathbb{V}(F) \subset \mathbb{P}_K^2$ it makes sense to talk about $D(L)$ for any L/K , with $D(K) \subset D(L)$.

Example 1.4.9. A projective line has the form $D = \mathbb{V}(aX + bY + cZ)$ for $a, b, c \in K$ not all zero, and where we may simultaneously scale a, b , and c . In particular, the set of all lines in $\mathbb{P}_K^2$ is itself a $\mathbb{P}^2(K)$, called the *dual projective plane*. (This duality has some lots of interesting consequences.) In general, this simultaneous scaling business feels awfully similar to what we have done above in the definition of projective spaces itself. This is no coincidence. The “space” of all curves of a given degree $d \in \mathbb{Z}_{\geq 0}$ is itself a projective space $\mathbb{P}^N(K)$ for some N (which?).

Remark 1.4.10. Doing the same procedure as above for arbitrary n gives us a *hypersurface* $D = \mathbb{V}(F) \subset \mathbb{P}_K^n$ associated to any nonzero homogenous $F \in K[X_0, \dots, X_n]$. When $n = 1$, for any field extension L/K , the set $D(L)$ is then finite and counts the number of roots—including those at infinity—of a (homogenized) single-variable equation in L (do you see why?). For example, if $K = \mathbb{Q}$ and $F = S^2 - 5ST + 5T^2$, then $D(\mathbb{Q}) = D(\mathbb{R}) = \emptyset$ but $D(\mathbb{C})$ consists of two points (c.f. 1.1.1(b)(i)).

Finally, as before we have the notions of homogenization and dehomogenization. We have that the locus $Z \neq 0$ (i.e., the complement of the line $\mathbb{V}(Z)$ at infinity) in $\mathbb{P}_K^2$ is naturally an $\mathbb{A}_K^2$:

$$\mathbb{A}_K^2 = \{Z \neq 0\} \subset \mathbb{P}_K^2.$$

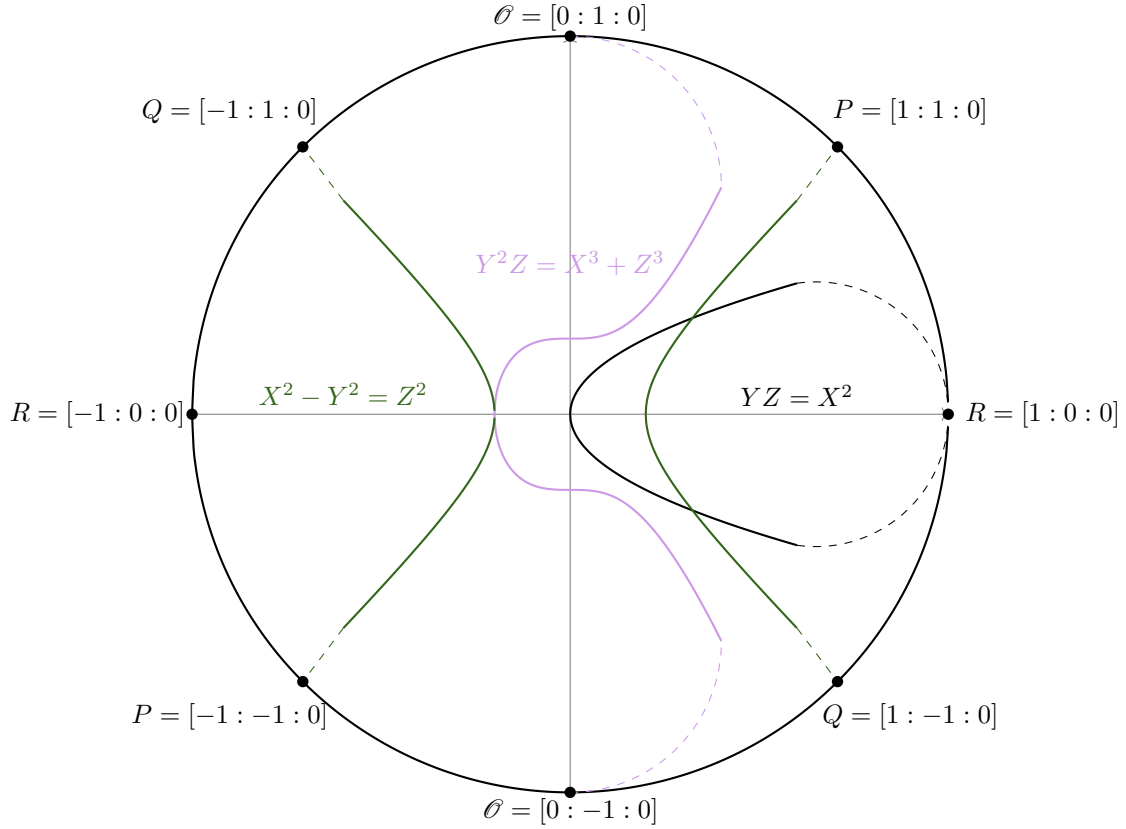


Figure 1.4.8: Some curves in the projective plane and their points at infinity.

For a projective curve $D = \mathbb{V}(F) \subset \mathbb{P}_K^2$, the intersection

$$C := D^\circ := D \cap \mathbb{A}_K^2 = \mathbb{V}(f) \subset \mathbb{A}_K^2,$$

where $f(x, y) := F(x, y, 1) \in K[x, y]$. This is the dehomogenization with respect to Z . Conversely, given an $f(x, y) \in K[x, y]$ of degree $d \in \mathbb{Z}_{\geq 1}$ and corresponding affine curve $C = \mathbb{V}(f)$, its *projective completion* is the projective curve

$$D := \overline{C} = \mathbb{V}(F) \subset \mathbb{P}_K^2,$$

where $F := Z^d f(X/Z, Y/Z) \in K[X, Y, Z]$. These two operations are basically inverses, but this is not literally true. For instance, we certainly have $(\overline{C})^\circ = C$ and $D \subset \overline{D^\circ}$, but equality need not hold in the latter—we may lose components corresponding to the line at infinity.

Exercise 1.4.11. What on earth does this last line mean? Are the operations of homogenization and dehomogenization inverses to each other?

However, in projective space the coordinates X, Y, Z are completely symmetrical. This flexibility of homogenization with respect to one variable, applying a linear transformation of coordinates, and then dehomogenization with respect to another yields lots of birational transformations of curves.

Example 1.4.12. Consider the affine curve $x^3 + y^3 = \alpha$ from 1.2.B. The transformation of coordinates given by

$$(x_1, y_1) := \left(\frac{12\alpha}{x+y}, \frac{36\alpha(x-y)}{x+y} \right)$$

is a birational transformation onto the curve $y_1^2 = x_1^3 - 432\alpha^2$. Of course, this arises from homogenization first to $X^3 + Y^3 = \alpha Z^3$, applying the transformation 1.2.7, and then dehomogenization with respect to Z_1 . (Make sure to convince yourself that this is a birational transformation!)

Here we used a linear change of coordinates in the middle. Let's study these a little more now.

Unimportant Remark 1.4.13. Sometimes there are equations “over $\mathbb{Z}$ ”, which can then be reduced mod p for various primes p . For instance, we could look at the curve C given by $Y^2Z = X^3 + Z^3$, or equivalently $y^2 = x^3 + 1$, and count points over $\mathbb{F}_p$ for various p . One would like to say that the original curve is defined over something like $\mathbb{Z}$ (i.e., $C \subset \mathbb{P}_{\mathbb{Z}}^2$) and these $\mathbb{F}_p$ -counts are the usual counts $C(\mathbb{F}_p)$ for the “extensions $\mathbb{F}_p/\mathbb{Z}$ ”. This is subtle, but can be made to work using the modern language of schemes. Indeed, figuring out the right definition of $\mathbb{P}_{\mathbb{Z}}^2$ (it may not be what you think!) was an impetus for the development of modern scheme theory.

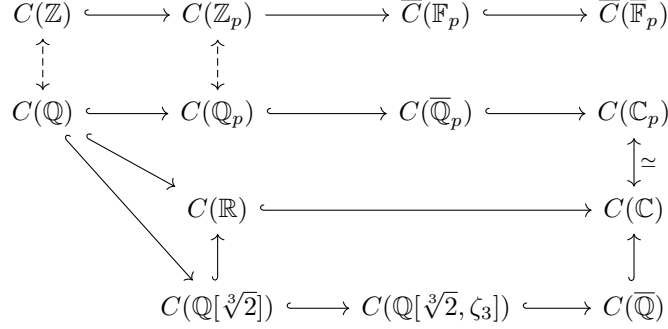


Figure 1.4.14: A figure explaining the relationship between the K -points of a projective curve C defined over $\mathbb{Q}$ (and lifted to $\mathbb{Z}$) for various K . Some of the terminology and relationships will be explained later.

1.4.B Projective Changes of Coordinates

One advantage of working with projective spaces is the presence of lots of extra symmetries not visible in the affine plane. These come from linear changes of coordinates and look like

$$(X_1, Y_1, Z_1) = (aX + bY + cZ, dX + eY + fZ, gX + hY + iZ) \quad (1.4.15)$$

for $a, b, c, d, e, f, g, h, i \in K$. Clearly, not all choices of these “transformation coordinates” will work: you certainly can’t take all of them to be zero. But, in fact the situation is a little stricter: you cannot take $(a, b, c, d, e, f, g, h, i) = (1, 0, 0, 1, 0, 0, 1, 0, 0)$ either. Similarly, but perhaps more subtly, you can’t take $(a, b, c, d, e, f, g, h, i) = (2, 3, 5, -1, 2, -2, 0, 7, -1)$ either (why?). To see exactly what this constraint is, it is helpful to write this change of coordinates in the form

$$\begin{bmatrix} X_1 \\ Y_1 \\ Z_1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}. \quad (1.4.16)$$

The idea is that if this transformation is to work, then there better not be a triple of numbers X, Y, Z , not all zero, such that $X_1 = Y_1 = Z_1 = 0$. This amounts to saying that the matrix A as above “representing this transformation” has no kernel, or equivalently (by a little linear algebra) has nonzero determinant. Conversely, if this matrix has nonzero determinant, then we get a way to invert this transformation by applying A^{-1} on the left. This is therefore the correct condition for invertibility: $\det A \neq 0$.

This matrix representation is also very helpful when we compose changes of coordinates: this just amounts to multiplying the corresponding change-of-coordinate matrices. This would lead us to believe that the group (not abelian!) of changes of coordinates in the group of invertible matrices, $\mathrm{GL}_3(K)$. But it’s a little more subtle than that. For instance, the change of coordinates $(X_1, Y_1, Z_1) = (\lambda X, \lambda Y, \lambda Z)$ for any $\lambda \in K^\times$ isn’t *really* a change of coordinates: it’s just the identity operation of $\mathbb{P}_K^2$. More generally, two matrices A and λA for $A \in \mathrm{GL}_3(K)$ and $\lambda \in K^\times$ represent the same change of coordinates. This tells us that the group of automorphisms is actually the quotient of $\mathrm{GL}_3(K)$ by its normal subgroup $K^\times$, which is called the *projective general linear group*

$$\mathrm{PGL}_3(K) := \mathrm{GL}_3(K)/K^\times.$$

Therefore, $\mathrm{PGL}_3(K)$ acts on $\mathbb{P}_K^2$ by linear changes of coordinates. (Similarly, $\mathrm{PGL}_{n+1}(K)$ acts on $\mathbb{P}_K^n$ for any $n \geq 1$. The action of $\mathrm{PGL}_2(K)$ on $\mathbb{P}_K^1$ is by *Möbius transformations*.)

Unimportant Remark 1.4.17. It can be shown (using heavy machinery that I will not get into here) that *all* (algebraic) automorphisms of $\mathbb{P}_K^2$ (over K) come from $\mathrm{PGL}_3(K)$: these are all the possible automorphisms you can have. Note that this is not true of $\mathbb{A}_K^2$: the map $(x, y) \mapsto (x, y - x^3)$ is a non-linear change of coordinates (i.e., automorphism) of $\mathbb{A}_K^2$.

Exercise 1.4.18. When does a change of coordinates preserve that line at infinity $Z = 0$? You will have to figure out what this means. For more along these lines, see ??.

The flexibility of these changes of coordinates, therefore, comes down to linear algebra. For instance, we can take any point to any other point by a linear change of coordinates. The same holds for any pair of points. The same is *not* true of any triple of points, since linear changes of coordinates preserve incidence relations: so you can't take a collinear triple of points to a non-collinear triple of points. But this is the only constraint: any collinear triple of points maps to any other, and any non-collinear triple maps to any other, by a linear change of coordinates. Finally, because projective changes of coordinates preserve incidence relations, this tells us that we can use them to map any line to any other. Similarly, any pair of lines goes to any other. Finally, not all triples of lines go to each other for the same reason as above: there is a difference between lines that “form a triangle” and coincident lines. Proving all of this amounts to a little linear algebra. You are invited to explore things of this sort in 2.1.13.

We have seen one example of change of coordinates in 1.2.B, and will see many examples of linear transformations throughout the course, so I don't feel terribly guilty giving too many examples now. Changes of coordinates are extremely useful, but lest you think those are all possible transformations, let me give you a warning:

Remark 1.4.19. Not every “change of coordinates” we will be interested in arises via a linear change of coordinates (on a projective closure). There are definitely other (birational) changes of coordinates we will be interested in. For a concrete example, see 2.1.12.

Chapter 2

Exercise Sheets

2.1 Exercise Sheet 1

2.1.A Numerical and Exploration

Exercise 2.1.1. Let $b, c \in \mathbb{R}$. When does the equation $x^3 + bx + c = 0$ have exactly one (resp. two, resp. three) real root(s)? (Hint: Play around on Desmos. A little calculus might come in handy.)

Exercise 2.1.2.

- (a) Let R be a UFD with fraction field K . State and prove the analog of the Rational Root Theorem (1.1.2) with $\mathbb{Z}$ replaced by R , and conclude that there is an algorithmic procedure (given the ability to find all factors of an element of R^1) for finding roots in K of single-variable polynomials with coefficients in K .
- (b) Using (a), or otherwise, determine all solutions in the field $\mathbb{Q}(i)$ to

$$2ix^3 + (2 - 4i)x^2 - (2 - 5i)x - (2 - i) = 0.$$

Exercise 2.1.3.

- (a) Find all integer solutions (x, y) to $3x + 5y + 4 = 0$ (and prove you have found all of them!).
- (b) Do the same for $4x + 6y + 5 = 0$.
- (c) For $a, b, c \in \mathbb{Z}$, give a necessary and sufficient criterion for the equation $ax + by + c = 0$ to have (i) infinitely many solutions, (ii) a finite nonzero number of solutions, and (iii) no solutions (x, y) in the integers.
- (d) Generalize (c) to an arbitrary number of variables. More precisely, given any $n \in \mathbb{Z}_{\geq 1}$ and integers $a_1, \dots, a_n, c \in \mathbb{Z}$, find necessary and sufficient conditions for the equation

$$a_1x_1 + \dots + a_nx_n + c = 0$$

to have integer solutions $(x_1, \dots, x_n)$ as in (i), (ii), and (iii).

Exercise 2.1.4.

- (a) Find all rational solutions (x, y) to $x^2 + 3y^2 = 2$.
- (b) Let p be a prime. Find all rational solutions (x, y) to the equation $x^3 + py^3 = p^2$.

Exercise 2.1.5.

- (a) Finish the computation started in class (1.1.6) to verify that the described procedure (“chord and tangent process”) yields Bachet’s formula 1.1.8.
- (b) In the case $c = 1$, why does Bachet’s formula get “stuck” at the points $(0, \pm 1)$?

Exercise 2.1.6.

- (a) Find all rational solutions (x, y) to $3x^2 - 2y^2 = 1$.
- (b) (**) Find all integer solutions (x, y) to $3x^2 - 2y^2 = 1$.

Exercise 2.1.7.

- (a) Find a prime at which the curve $x^2 + y^2 = 3$ is locally obstructed.
- (b) (*) Find *all* primes at which the curve $x^2 + y^2 = 3$ is locally obstructed.

Exercise 2.1.8. (Adapted from [1, Exercise 2.4].)

- (a) Consider the plane curve $y^2 = x^3 + 1$. For each prime p , let n_p be the number of solutions (x, y) to this equation modulo p (i.e., the number of $\mathbb{F}_p$ -points on this affine curve). Using your favorite computer software, compute the values n_p for all primes $p \leq 100$. Do you see any patterns? Make and prove conjectures.
- (b) Repeat the same procedure for the curve $y^2 = x^3 + x$.

Exercise 2.1.9 (Fisher). Let E be the elliptic curve over $\mathbb{Q}$ defined by the equation $y^2 + y = x^3 - x$.

- (a) Draw a graph of the real points of E .

¹This is the same as having the ability to find *one* factorization of any element of R , along with the knowledge of all units of R .

- (b) Let $P := (0, 0) \in E(\mathbb{Q})$. Get Sage to compute nP for $n = 0, 1, \dots, 20$. What do you notice about the denominators? Make and prove conjectures. (Hint: Can you express the operation of adding P formulaically?)

2.1.B PODASIPs

Prove or disprove and salvage if possible the following statements.

Exercise 2.1.10. (Hasse-Minkowski for Quadratic Equations) Let $b, c \in \mathbb{Q}$. The quadratic equation $x^2 + bx + c = 0$ has a root in $\mathbb{Q}$ iff it has a root in $\mathbb{Q}_p$ for all primes $p \leq \infty$. (For those not too familiar with $\mathbb{Q}_p$ yet, the only fact you will need is: if a $d \in \mathbb{Q}^\times$ is a square in $\mathbb{Q}_p$, then $v_p(d)$ is even. You may assume this for now.)

Exercise 2.1.11 (Homogeneous Polynomials). Let K be a field, $d \in \mathbb{Z}_{\geq 0}$, and $F \in K[X, Y, Z]$. The following conditions are equivalent:

- (a) Every monomial in F has total degree d .
- (b) We have the equality $F(WX, WY, WZ) = W^d F(X, Y, Z)$ in the polynomial ring $K[W, X, Y, Z]$.
- (c) For all $\lambda \in K$, we have $F(\lambda X, \lambda Y, \lambda Z) = \lambda^d F(X, Y, Z)$.
- (d) We have that

$$X \frac{\partial F}{\partial X} + Y \frac{\partial F}{\partial Y} + Z \frac{\partial F}{\partial Z} = d \cdot F.$$

Exercise 2.1.12. (Adapted from [1, Exercise 1.15].) Let K be a field, $g(u) \in K[u]$ be a monic quartic polynomial, and consider the curve

$$v^2 = g(u).$$

- (a) Suppose we are given a root $\alpha \in K$ of g , and a $\beta \in K^\times$. The transformation

$$(x, y) := \left(\frac{\beta}{u - \alpha}, \frac{\beta^2 v}{(u - \alpha)^2} \right)$$

transforms $v^2 = g(u)$ to a curve of the form

$$y^2 = f(x)$$

for some cubic polynomial $f(x) \in K[x]$. This transformation is birational (i.e., a one-to-one correspondence between points of the two curves, except possibly for a finite number of points). Note that this birational transformation does not arise from a linear change of coordinates on (projective closures of) these curves.

- (b) If $X, Y, Z \in K$ satisfy $X^4 + Y^4 = Z^4$ with $Z \neq X$, then

$$(x, y) := \left(2 \left(\frac{Z + X}{Z - X} \right), \frac{4Y^2}{(Z - X)^2} \right)$$

lies on the curve

$$y^2 = x^3 + bx$$

for some $b \in K$. This b is unique. (What is it?)

(Hint: You will need some condition on g for (a) to work. What is this condition? What implications does this condition have for the roots of f ? What do parts (a) and (b) have to do with each other? Part (b) is definitely not correct as stated.)

Exercise 2.1.13. Let K be a field.

- (a) Any two lines in $\mathbb{P}^2(K)$ meet in exactly one point. Any two points in $\mathbb{P}^2(K)$ simultaneously lie on exactly one line.
- (b) Given any two ordered quadruples (P_1, P_2, P_3, P_4) and (Q_1, Q_2, Q_3, Q_4) of points of $\mathbb{P}^2(K)$, there is a unique linear change of coordinates φ such that $\varphi(P_i) = Q_i$ for $i = 1, 2, 3, 4$. The same result holds for ordered pentuples.

Exercise 2.1.14 (Fisher). Let $C \subset \mathbb{P}_{\mathbb{Q}}^2$ be a smooth cubic curve. If $K/\mathbb{Q}$ is a quadratic or cubic number field, then $C(K) \neq \emptyset$ implies that $C(\mathbb{Q}) \neq \emptyset$.

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